The Extremal Principle — problem 3.25

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Consider a walk in the plane according to the following rules. From a given point $P(x, y)$ we may move in one step to one of the four points $U(x, y+2x)$, $D(x, y-2x)$, $L(x-2y, y)$, $R(x+2y, y)$ with the restriction that we cannot retrace a step we just made. Prove that, if we start from the point $(1, \sqrt{2})$, we cannot return to this point any more (HMO 1990).